



Ahmed Aziz Ben Haj Hmida

Machine Learning Engineer, graduate of the École Polytechnique Fédérale de Lausanne (EPFL), specializing in Computer Vision, Large Language Models, and scalable AI systems.

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Education

EPFL – École Polytechnique Fédérale de Lausanne

M.Sc. in Data Science

GPA: 5.3/6

Relevant Coursework: Machine Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Data Analysis, Database Systems, Data Science for Business

Lausanne, Switzerland

Sep 2023 – Mar 2026

Technical University of Denmark (DTU)

3rd Year of Bachelor - Exchange

GPA: 5.35/6

Lyngby, Denmark

Aug 2022 – Jul 2023

EPFL – École Polytechnique Fédérale de Lausanne

B.Sc. in Communication Systems

GPA: 5.1/6

Lausanne, Switzerland

Sep 2020 – Jul 2023

Pioneer School of Nabeul

Baccalaureate in Mathematics

Nabeul, Tunisia

Sep 2018 – Jun 2019

Professional Experience

Rollomatic SA

Master's Thesis – Curve-Based CNC Tool Similarity Retrieval - Grade 6/6

Le Landeron, Switzerland

Aug 2025 – Mar 2026

- Engineered an end-to-end similarity search system using geometric representations and deep visual embeddings.
- Built scalable indexing and evaluation pipelines for industrial workflows.
- Optimized for industrial workflows using rigorous testability to ensure seamless R&D integration.
- Improved retrieval accuracy by **11%** and reduced latency by **95–99%**.

Supervisors: Prof. Jean-Philippe Thiran - jean-philippe.thiran@epfl.ch, Emna Maghrebi - e.maghrebi@rollomatic.ch

Rollomatic SA

Machine Learning Internship

Le Landeron, Switzerland

Feb 2025 – Aug 2025

- Developed hybrid retrieval pipelines combining structured parameters and visual embeddings.
- Preprocessed industrial datasets and optimized similarity learning workflows.

Supervisor: Emna Maghrebi - e.maghrebi@rollomatic.ch

Research Projects

Unsupervised Reverse Knowledge Distillation and Model Merging 🌀

Aug 2024 – Feb 2025

EPFL LTS4 Lab

- Finetuned compact specialized models and transferred their knowledge into larger complex architectures.
- Applied model merging techniques to preserve general capabilities while improving niche-task performance.

Supervisor: Prof. Pascal Frossard - pascal.frossard@epfl.ch, Ortal Senouf

Discovering promising new startups through evaluating teams' potentials 🌀

Sep 2023 – Dec 2023

EPFL ENTC Lab

- Used ML methods to infer speakers' MBTI personalities and emotions from entrepreneurial discussions.
- Built predictive models to identify promising teams to create successful startups.

Supervisor: Prof. Davide Bavato

Technical Skills

Languages: Python, Java, SQL, MATLAB, Scala

ML & AI: NumPy, Pandas, scikit-learn, PyTorch, PySpark

Computer Vision: OpenCV, scikit-image, torchvision, Vision Transformers, CNNs, ResNet

Natural Language Processing: spaCy, HuggingFace, Quantization, LoRA, RLHF, PEFT, SentenceTransformers

Agentic AI: LangChain, LangGraph, LlamaIndex, Vector Search, FAISS, PostgreSQL, Prompt Engineering

DevOps & MLOps: Git, Docker, Kubernetes, FastAPI, MLflow, W&B, DVC, GitHub Actions, GCP, AWS, Streamlit

Visualization: Matplotlib, Seaborn, Plotly, Power BI

Personal & Academic Projects

- LLM Agent–Driven Car Search Assistant**  Apr 2026 - May 2026
- Built a Streamlit frontend where users describe a car in chat and LangGraph with LangChain and Ollama fills a structured profile and queries PostgreSQL.
 - Displayed ranked listing cars with match scores plus Plotly and Matplotlib charts, generated PDFs for single searches and for comparing two or three saved car profiles, and loaded the listings into Postgres.
- LLM Teaching Assistant for MCQ Answering**  Feb 2024 - Jun 2024
- Fine-tuned Qwen1.5 using SFT and DPO with RAG and 4-bit quantization for efficient response generation.
 - Developed a chatbot User Interface and a robust evaluation pipeline using LLM-as-a-Judge methods and cross-validation to ensure pedagogical accuracy.
- Coin Detector and Counter**  Feb 2024 - Jun 2024
- Built segmentation and classification pipelines using CNNs, ViTs, and ResNet50.
 - Detected and identify coins accurately across diverse backgrounds.
- RL for Motor Skill Transfer**  Feb 2024 - Jun 2024
EPFL Alexander Mathis Lab
- Trained an expert model on the Myosuite baoding balls task and distilled its motor strategies to novice agents.
 - Improved learning efficiency and performance of novice agents in complex motor control tasks.
- Machine Learning Operations**  Jan 2023 - Mar 2023
- Built scalable ML pipelines using PyTorch, Docker, Kubernetes, MLflow, DVC, AWS, and GCP.

Certifications

- Structuring Machine Learning Projects** – DeepLearning.AI Sep 2023 - Oct 2023
Certificate in Artificial Intelligence – GOMYCODE Jun 2023 - Sep 2023

Leadership and Volunteering

- EPFL Coaching** – CMS section Coach Sep 2023 - Jun 2024
Rotary International – Member Jul 2018 - Apr 2019
Tunisian Red Crescent – Volunteer May 2018 - Apr 2019

Languages

Arabic (Native), French (Native/C2), English (C1), German (A2)